

SALIVARY GLAND NEOPLASMS

VOLUME V

Adjuvant Therapy, Recurrent Disease, and Surveillance

From final pathology to radiation, salvage, precision oncology, and lifelong follow-up



Figure 1. Original synthesis. Surgery changes the anatomy but does not erase tumor biology. The postoperative plan should remain spatial, histology-specific, and explicit about treatment triggers.

THE CENTRAL QUESTION | After the operation, which failure pattern is most likely - and what treatment, surveillance, or systemic strategy can meaningfully change it?

CHAPTER 0

Orientation

How to convert an operation and pathology report into an active oncologic plan.

Salivary cancer follow-up is not a generic head-and-neck template. It must be adapted to the operation, histology, nerve pathways, nodal pattern, metastatic tempo, and actionable biology.

The operation ends with altered anatomy and a pathology report. The next task is to convert those records into a forecast: where could microscopic disease remain, which failure would matter first, and what intervention has enough evidence to justify its morbidity? A high-grade salivary duct carcinoma may progress rapidly in nodes and viscera. Adenoid cystic carcinoma (ACC) may remain locally controlled yet develop slowly enlarging pulmonary metastases years later. Secretory carcinoma can change from an uncommon chemotherapy-resistant diagnosis to a highly targetable disease when a true NTRK fusion is confirmed.

Learning objectives

- Interpret final pathology as a local-bed, neural, regional, distant, and molecular failure map.
 - Identify strong and discretionary PORT indications and explain guideline differences.
 - Reconstruct postoperative targets from operative anatomy rather than from the incision alone.
 - Explain why concurrent chemotherapy is not routinely added to salivary RT outside a trial.
- Re-stage locoregional or distant recurrence before labeling it incurable.
 - Choose observation, local ablation, cytotoxic therapy, targeted therapy, or a trial according to tempo and goal.
 - Use AR, HER2, NTRK, RNA fusion testing, and broad sequencing without overinterpreting weak signals.
 - Integrate dental, swallowing, facial, ocular, auditory, neurologic, and psychosocial care into surveillance.

Contents

Chapter	Core question
1. Final pathology	What pattern of failure does the resection specimen predict?
2. PORT	Who should receive postoperative radiation, and what does it accomplish?
3. Target design	Which bed, nerve pathways, and nodal echelons must be covered?
4. Definitive RT	How are unresectable disease and the uncertain role of cisplatin approached?
5. Recurrence	Can salvage surgery, local therapy, or reirradiation still produce durable control?
6. Distant disease	When should metastases be observed, ablated, or treated systemically?
7-8. Systemic therapy	Which histology, biomarkers, tempo, and toxicity determine the next drug?
9. Surveillance	How should examination and imaging change over time?
10. Survivorship	Which surgical, radiation, and systemic consequences require active care?
11-12. Cases and workflow	Can the framework produce a defensible tumor-board and oral-board plan?

EVIDENCE CALIBRATION | ASCO and ESMO-EURACAN are the principal management anchors. Many systemic data derive from small single-arm studies. Response rates should orient discussion, not be treated as head-to-head comparisons. Emerging 2026 evidence is labeled separately from established practice.

CHAPTER 1

The Final Pathology Conference

The resection specimen predicts where failure may occur and what the next treatment must cover.

The pathology report is the handoff from resection to risk-adapted care. It should be read beside the preoperative imaging, operative note, specimen orientation, and nerve map.

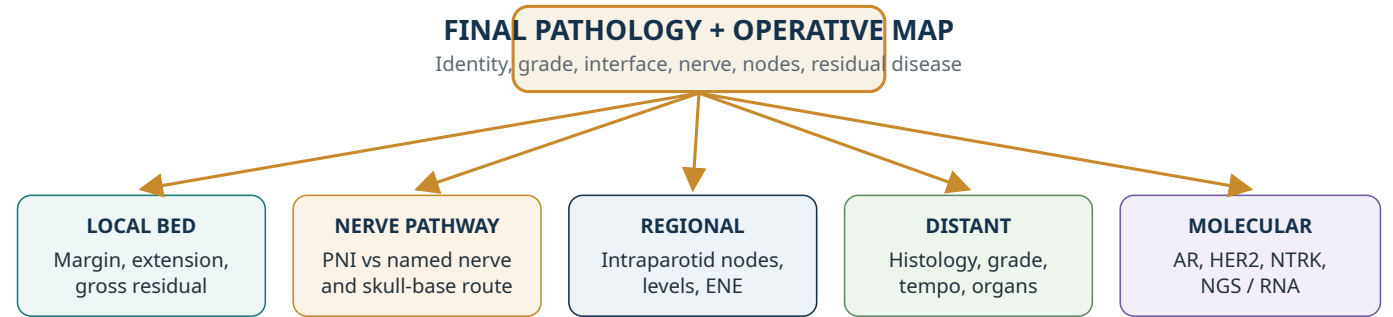


Figure 2. Original synthesis. Adverse features are not interchangeable; each predicts a different spatial or temporal pattern of failure.

1.1 The minimum postoperative data set

Domain	Resident questions	Management consequence
Histology and grade	Is the entity definitive? Is there a high-grade component or transformation?	Changes PORT strength, nodal/distant risk, molecular workup, and surveillance intensity.
Primary and extent	Exact site, size, adjacent-structure extension, T category, and intentional residual?	Defines the bed, barriers crossed, and probability of microscopic residual disease.
Margins	Positive, close, or widely clear? Which mapped patient-side interface? Is re-resection feasible?	Positive margin strongly supports PORT; a focal accessible margin may justify revision first.
Neural disease	Incidental microscopic PNI, large/named nerve involvement, symptoms, or gross spread?	Determines MRI review, nerve-pathway target, skull-base coverage, and surveillance signs.
LVI and nodes	LVI, intraparotid nodes, levels, number/size, ENE, and violated nodes?	Defines high-dose nodal volumes and next-echelon elective treatment.
Molecular phenotype	AR, HER2, NTRK fusion, or other alteration relevant to advanced therapy?	Usually does not replace local therapy, but preserves options for future measurable disease.

CLINICAL BRIDGE | A positive intraparotid node, a positive facial-nerve interface, and incidental microscopic PNI are all “high risk,” but they do not create the same target or surveillance plan.

CHAPTER 1 CONTINUED

From pathology language to a spatial postoperative plan

1.2 Re-resection before radiation

A positive margin is not automatically solved by radiation. If the involved surface is anatomically identifiable and can be cleared with acceptable morbidity, re-resection should be discussed before PORT. The decision depends on whether the edge represents avoidable residual disease or an intentionally close boundary at a critical structure, whether wider surgery meaningfully improves control, and whether revision would unacceptably delay adjuvant treatment. Patients undergoing salvage surgery for recurrent salivary cancer should again be evaluated for adjuvant therapy.^[1]

BOARD PEARL | “Close” is not a universal numeric diagnosis across salivary sites. Interpret the measured distance with histology, infiltrative pattern, anatomic barrier, specimen handling, and whether the sample is a true patient-side margin.

1.3 Pathology features are spatial instructions

ACC and other neurally tropic tumors require attention to a named nerve pathway even when the main specimen margin is negative. Nodal metastasis requires more than treating the primary bed: involved levels, intraparotid nodes, extranodal extension, and the next echelon at risk must be integrated. High-grade transformation means that the aggressive component should drive regional and distant risk assessment. Conversely, an early low-grade carcinoma removed in an intact specimen with a clearly mapped negative margin may be observed rather than irradiated reflexively.

1.4 The operative-to-radiation handoff

PRIMARY Site, T category, structures removed or preserved, reconstruction	MARGINS Named patient-side interfaces; whether a close boundary was intentional	NERVES Symptoms, branches sacrificed, proximal/distal margins, skull-base concern
NODES Intraparotid vs cervical, levels, ENE, violated nodal disease	RESIDUAL Gross, microscopic, uncertain postoperative change, or none	TISSUE Blocks/slides available for AR, HER2, NTRK, NGS, and RNA testing

Chapter synthesis

- Review pathology beside imaging, operative anatomy, and specimen orientation.
- Margins, grade, PNI, LVI, T category, and nodal disease predict different failure pathways.
- Clear a safely correctable positive margin rather than assuming PORT can compensate for avoidable disease.
- Preserve adequate tissue and testing pathways for recurrence, but do not let molecular testing substitute for local-regional control.

CHAPTER 2

Postoperative Radiation Therapy

Start with the adverse feature, then design the volume it creates.

PORT is treatment of microscopic disease. Its indication and its volume should both be explicit.

ASCO recommends PORT for high-grade tumors, positive margins, PNI, nodal metastases, lymphatic or vascular invasion, and T3-T4 disease. PORT may be offered for close margins or intermediate-grade tumors, where evidence is weaker.^[1]

2.1 Strong versus individualized indications

Feature	PORT stance	Why it matters
High-grade histology or high-grade transformation	Offer PORT	Higher local-regional risk; distant risk remains histology-dependent.
Positive margin	Offer PORT; discuss re-resection first	Microscopic or gross residual disease is a dominant local-risk feature.
Perineural invasion	Offer PORT	Failure may extend beyond the apparent bed along the associated nerve.
Pathologic nodal metastasis	Offer PORT to appropriate bed and nodal levels	Regional risk persists after dissection, especially with multiple nodes or ENE.
Lymphatic or vascular invasion	Offer PORT	Marker of spread beyond the gross tumor.
T3-T4 disease	Offer PORT	Large size or adjacent-structure extension raises microscopic residual risk.
Close margin or intermediate grade	Individualize	Evidence is less definitive; anatomy and coexisting adverse features determine the balance.
T1-T2 low-grade, clear margin, no adverse feature	Observation is often appropriate	Incremental control benefit may not justify RT morbidity after adequate surgery.

GUIDELINE DIFFERENCE - ACC | ASCO recommends PORT after resection of all ACC. ESMO uses a more feature-directed framework, emphasizing T3-T4 disease, close/incomplete margins, intermediate/high grade, and perineural growth. In a small completely resected early ACC, state which guideline philosophy is being followed and discuss neural route, site, margin confidence, and morbidity rather than presenting the choice as universally settled.

2.2 What PORT accomplishes

The most consistent benefit is improved local-regional control. Overall-survival benefit is harder to define because salivary histologies are heterogeneous, distant failure may dominate, and retrospective cohorts are confounded by selection. PORT should not be dismissed because randomized data are sparse, but neither should it be presented as a way to eliminate late pulmonary or visceral metastasis.

COUNSELING LANGUAGE | “Radiation is recommended because pathology shows a meaningful risk that microscopic disease remains in the bed, nerve pathway, or nodes. Its principal purpose is local-regional control; it cannot remove all future distant risk.”

CHAPTER 2 CONTINUED

Timing, dose hierarchy, and elective nodal treatment

2.3 Timing and postoperative readiness

ASCO recommends initiating PORT within eight weeks of surgery.^[1] This is a planning deadline, not a reason to irradiate an unhealed wound. Use the interval actively: finalize pathology, arrange radiation consultation, obtain dental evaluation when teeth or mandible will receive dose, establish nutrition and swallowing support, document hearing when cochlear dose or ototoxic therapy is relevant, manage infection or fistula, and complete simulation. Preventable referral and dental delays should not be normalized.

2.4 Dose is tied to residual-risk tier

Volume	Conceptual dose range	Clinical caveat
Elective nodal or named-nerve pathway	Approximately 46-54 Gy	Microscopic-risk tissue; ASCO specifically describes 46-54 Gy along an involved nerve toward the skull base as reasonable.
High-risk postoperative bed	At least 60 Gy	Conventionally fractionated high-dose target after resection.
Positive-margin / very-high-risk interface	Often escalated above 60 Gy	Exact dose depends on residual probability and organs at risk.
Gross residual / definitive disease	Approximately 66-70 Gy-equivalent	Curative gross-disease strategy; geometry and modality matter.
Reirradiation	No generic range	Requires composite dose, tissue recovery assumptions, target size, and patient-specific benefit.

These are conventional planning concepts, not prescriptions. Fractionation, residual disease, particle modality, prior radiation, and nearby normal structures can change the final plan.

2.5 Elective neck irradiation

When the cN0 neck was not surgically treated, elective nodal irradiation may be offered for T3-T4 cancers or high-grade malignancies. After a positive neck dissection, involved levels receive high-dose treatment and the next echelon may receive elective dose. The exact levels follow the primary site, prior dissection, and nodal distribution rather than a generic parotid template.^{[1][2][3]}

BEFORE SIMULATION

- Wound and flap readiness
- Dental and fluoride plan
- Nutrition and swallowing baseline
- Eye closure and facial function
- Hearing baseline when relevant

WHAT PORT CANNOT DO

- Correct avoidable gross residual disease
- Erase late ACC pulmonary risk
- Substitute for a defined neck plan
- Repair missing nerve or swallow rehabilitation

Chapter synthesis

- Offer PORT for established adverse features; individualize close margins and intermediate grade.
- Discuss the ASCO-ESMO difference for completely resected early ACC.
- Start actively planning early enough to begin by eight weeks when clinically ready.
- State the risk tier, target, dose concept, and expected benefit rather than saying only “adjuvant RT.”

CHAPTER 3

Radiation Target Design

Reconstruct the disease in three dimensions before drawing a target.

The target is a reconstructed anatomic story: surgical bed, residual-risk interfaces, nerve pathways, nodal echelons, and organs at risk.

TARGETS FOLLOW FAILURE PATHWAYS - NOT THE OPERATIVE INCISION

1 RECONSTRUCT Preoperative imaging + operative report + specimen map + pathology	2 TIER RISK Gross/high-dose, intermediate/elective, or observation	3 TRACE ROUTES Named nerve to skull base and nodal next echelon	4 PROTECT FUNCTION Cochlea, brainstem, temporal lobe, mandible, oral cavity, swallowing structures
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Figure 3. Original synthesis. A generic preauricular field can miss a deep-lobe interface, stylomastoid-foramen route, palatal nerve pathway, or undissected nodal basin.

3.1 High-dose and elective targets

Risk pattern	Volume concept	Resident contribution
Positive or close primary margin	Map the specific anatomic interface and any re-resection; avoid the unhelpful label “deep.”	Orient the specimen and identify the true patient-side surface in the operative note.
Perineural disease	Cover the associated nerve toward the relevant skull-base foramen when indicated.	Document symptoms, branch involvement, sacrificed segments, and proximal/distal margins.
Pathologic nodes	High dose to involved levels and nodal bed; elective dose to adjacent echelons at risk.	Separate intraparotid from cervical nodes; record levels, ENE, and any violated node.
Minor-gland primary	Target follows mucosal-site resection, bone interfaces, and site-specific neural/lymphatic anatomy.	Provide exact subsite, mucosal margins, bone involvement, and reconstructive boundaries.
Gross residual / unresectable disease	Gross-disease target plus appropriate neural and nodal routes.	Clarify what is truly residual versus expected postoperative enhancement.

RESIDENT TAKEAWAY | The radiation consultation improves when the surgeon provides a spatial report: tumor interface, nerve course, true margins, nodal levels, reconstruction, and intentional residual disease.

CHAPTER 3 CONTINUED

Named-nerve pathways

The pathway matters because neural failure can occur beyond the visible primary bed.

Microscopic PNI is a pathology finding; clinical or radiographic perineural spread is a disease distribution. Gross named-nerve disease may produce pain, numbness, weakness, foraminal enlargement, denervation change, or enhancement along the nerve and requires a larger high-risk map.

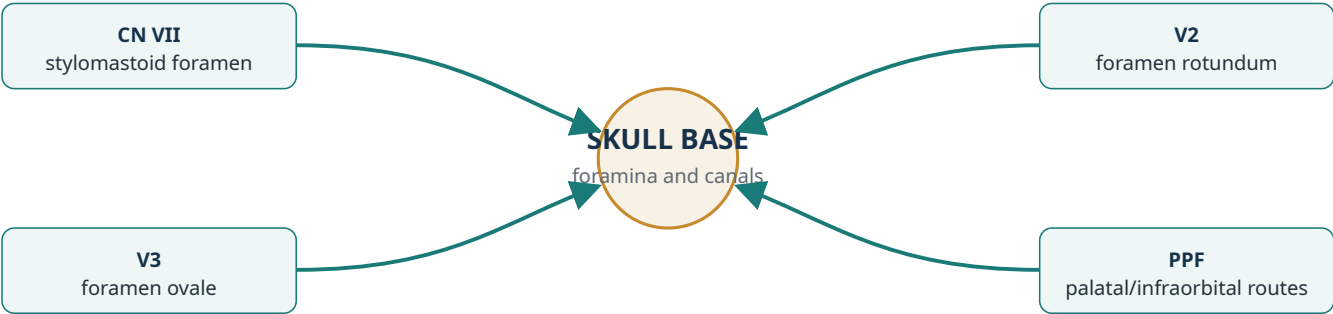


Figure 4. Original schematic. The target is traced retrograde from the involved peripheral nerve toward the relevant skull-base corridor; the exact extent depends on clinical and radiographic disease.

3.2 Common neural routes

Primary context	Peripheral route	Proximal corridor	Clinical/radiographic clue
Parotid / CN VII	Facial nerve trunk or branch	Stylomastoid foramen -> mastoid facial canal -> geniculate region	New facial weakness, twitching, otalgia, mastoid-foramen enhancement
Auriculotemporal route	Auriculotemporal nerve to V3	Infratemporal fossa -> foramen ovale	Preauricular pain, V3 sensory change, masticator denervation
Hard/soft palate	Greater or lesser palatine nerves	Pterygopalatine fossa -> V2 -> foramen rotundum	Palatal numbness, V2 symptoms, pterygopalatine-fossa enhancement
Upper lip / midface	Infraorbital nerve	Infraorbital canal -> V2 -> foramen rotundum	Cheek or upper-lip numbness, infraorbital-canal change
Floor of mouth / tongue	Lingual nerve	V3 -> foramen ovale	Tongue/floor-of-mouth sensory change, deep pain
Lower lip / mandible	Mental -> inferior alveolar nerve	Mandibular canal -> V3 -> foramen ovale	Numb-chin syndrome, mandibular-canal enlargement

DIAGNOSTIC TRAP | Do not convert every microscopic PNI focus into automatic full skull-base high-dose coverage, and do not treat gross radiographic nerve spread as a focal gland-bed problem. Review the actual nerve caliber, symptoms, margin, and MRI with neuroradiology and radiation oncology.

CHAPTER 3 CONTINUED

Imaging, modality selection, and toxicity

3.3 Perineural spread on MRI

Contrast-enhanced fat-suppressed MRI is the principal study for neural mapping. Review enhancement, caliber change, foraminal asymmetry, denervation, and continuity from the primary bed toward the skull base.

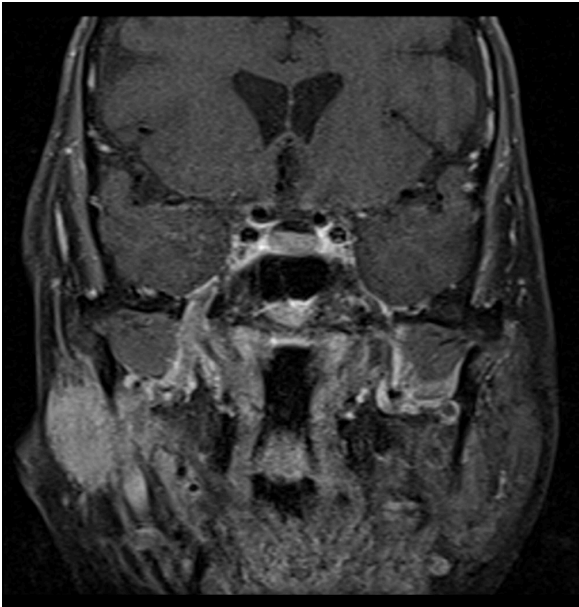


Figure 5. Open-license MRI example of facial-nerve perineural tumor spread. Jto410, CC BY-SA 3.0, via Wikimedia Commons. Use the image with the operative nerve map and a named-nerve schematic rather than relying on a generic “PNI present” label.

3.4 Particle therapy in context

Photon IMRT remains standard and widely available. Proton, neutron, and carbon-ion approaches may be considered for selected skull-base or ACC presentations, but ASCO found no disease-wide indication requiring heavy particles over photons or electrons. ESMO notes potential advantages in selected cases while emphasizing nonrandomized evidence and limited availability.^{[1][2]}

PRACTICAL QUESTION | Can the modality cover the true target while reducing a clinically meaningful dose to cochlea, temporal lobe, brainstem, optic structures, mandible, oral cavity, or swallowing structures?

3.5 Acute and late toxicity

Time	Common toxicity	Prevention / management priorities
During RT	Dermatitis, mucositis, taste loss, xerostomia, fatigue, odynophagia, nausea, weight loss	Skin/oral care, analgesia, nutrition, swallowing exercise, hydration, adherence
Early months	Persistent xerostomia, thick secretions, trismus, dysphagia, lymphedema, neuropathic pain	Dental prevention, jaw stretching, SLP, lymphedema therapy, pain-directed treatment
Late	Caries/osteoradionecrosis, fibrosis, trismus, cranial neuropathy, hearing loss, temporal-lobe or brainstem injury	Lifelong fluoride/dental care, rehabilitation, audiology, MRI for new neurologic symptoms
Reirradiation	Soft-tissue necrosis, carotid blowout, bone injury, cranial neuropathy, severe fibrosis	Composite-dose review, strict selection, experienced planning, frank counseling

Chapter synthesis

- Target design begins with preoperative imaging, operative anatomy, specimen orientation, and pathology.
- Trace named-nerve risk to the correct skull-base corridor; distinguish microscopic PNI from gross spread.
- Particle therapy is a selected dosimetric tool, not a universal salivary-cancer requirement.
- Functional preparation and toxicity prevention begin before simulation and continue lifelong.

CHAPTER 4

Unresectable Disease and the Role of Chemotherapy

Definitive RT is established; routine salivary chemoradiation remains unproven.

4.1 What “unresectable” should mean

Unresectability may reflect extension that prevents a complete operation, unacceptable neurologic or vascular morbidity, inability to reconstruct a safe defect, or medical comorbidity that makes surgery prohibitive. The term should not be used loosely for a difficult but potentially curable skull-base or parapharyngeal case without expert multidisciplinary review. Distant metastases also do not make the primary irrelevant when local progression threatens pain, bleeding, airway, cranial nerves, or skin.

4.2 Definitive radiation

For patients who cannot undergo resection, ASCO recommends radiotherapy to the gross primary and appropriate nodal disease. Conventional definitive plans commonly deliver approximately 66-70 Gy-equivalent to gross disease, with neural and elective nodal routes selected by histology and anatomy.^{[1][2]}

CONCURRENT CHEMOTHERAPY: DO NOT IMPORT A SQUAMOUS-CELL RULE WITHOUT EVIDENCE

1 KNOWN PORT improves local-regional control in adverse-feature disease.	2 UNKNOWN Does adding weekly cisplatin improve outcomes enough to justify toxicity?	3 CURRENT PRACTICE Do not routinely add concurrent chemotherapy outside a trial.	4 PENDING EVIDENCE NRG/RTOG 1008 primary completion is estimated October 31, 2026; no results posted as of July 29, 2026.
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Figure 6. NRG/RTOG 1008 enrolled 252 patients and compares postoperative RT alone with RT plus weekly cisplatin. Eligibility is restricted to defined high-risk histologies and features, so even a positive trial would require attention to generalizability.^[4]

4.3 Current stance

Situation	Current stance	Reason
Resected high-risk disease receiving PORT	Do not routinely add concurrent chemotherapy outside a trial	No mature randomized evidence; acute toxicity is expected to increase.
Unresectable disease receiving definitive RT	Do not routinely add concurrent chemotherapy outside a trial	Evidence consists mainly of small series and extrapolation.
Adjuvant AR- or HER2-directed therapy after complete local treatment	Not routine outside a trial	Activity in metastatic disease does not establish benefit against microscopic residual risk.
Gross recurrent/metastatic disease	Use histology- and biomarker-directed systemic therapy when treatment is indicated	Here the drug treats measurable disease with a defined clinical goal.

BOARD TRAP | Positive margins and extranodal extension establish concurrent cisplatin in selected mucosal squamous-cell carcinomas. That standard should not be automatically transferred to salivary carcinoma.

CHAPTER 5

Locoregional Recurrence and Salvage

Confirm, map, classify, and then choose the most effective local or systemic strategy.

Recurrence should trigger a new diagnostic and anatomic workup, not a reflex declaration of incurability.



Figure 7. Original synthesis. The key questions are not simply “Has it returned?” but “Where, how fast, with which biology, and can all meaningful disease be controlled?”

5.1 Restaging the recurrence

- **Local and neural map:** contrast MRI for the bed, deep spaces, skull base, and perineural pathways when clinically plausible.
- **Bone, nodes, and chest:** contrast CT for cortical bone, nodal anatomy, thorax, or urgent anatomic definition; chest CT is important because lung disease may coexist.
- **Whole-body staging:** PET/CT selectively for high-grade recurrence, uncertain extent, or when additional sites would change salvage intent.
- **Tissue:** biopsy the safest informative site and preserve material for AR, HER2, NTRK, DNA/RNA sequencing, and exclusion of a metastatic mimic.
- **Prior treatment:** obtain the original radiation plan and cumulative dose before discussing reirradiation.

5.2 Salvage surgery

Surgery offers the clearest chance of durable control for isolated, resectable locoregional recurrence, particularly after a long disease-free interval and when complete resection is achievable. Scar obscures planes, and recurrence may track along a nerve or reconstructive interface. Reassess facial and lower cranial nerve reconstruction, skull-base access, carotid involvement, wound coverage, and the possibility of postoperative RT or reirradiation. ASCO recommends evaluating revision-surgery patients for potential adjuvant treatment.^[1]

FEATURES FAVORING SALVAGE • Isolated or limited recurrence • Longer disease-free interval • Complete resection plausible • Acceptable neurologic and vascular cost • Reconstructable defect	FEATURES AGAINST CURATIVE SALVAGE • Macroscopic residual disease expected • Uncontrolled distant progression • Carotid or skull-base geometry precludes control • Severe prior-treatment morbidity • No meaningful functional or symptom benefit
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CHAPTER 5 CONTINUED

Reirradiation and focal local therapy

5.3 Reirradiation is not a single treatment

The word “reirradiation” should trigger a composite-dose and normal-tissue analysis. The team must know the prior dose distribution, interval, current target size, and which structure will limit treatment - carotid, mandible, temporal lobe, brainstem, optic apparatus, skin, flap, or cranial nerve.

Clinical setting	Potential strategy	What must be demonstrated
RT-naïve recurrence after previous surgery	Definitive RT or postoperative RT after salvage	Treat according to current gross/microscopic risk; not technically “reirradiation.”
Salvage surgery after prior RT	Postoperative reirradiation for selected very-high-risk pathology	Requires expected benefit beyond surgery and acceptable composite dose.
Unresectable focal recurrence in prior field	Definitive reirradiation, sometimes stereotactic/hypofractionated	Small target, safe geometry, long interval, and meaningful control horizon favor use.
Oligometastatic or oligoprogressive site	SBRT/ablation or surgery	Treat all active sites or a limited resistant site while systemic therapy controls the remainder.
Diffuse local plus systemic progression	Palliative local RT plus systemic/palliative strategy	Urgency is defined by pain, bleeding, airway, neurologic or organ threat.

5.4 Four questions before reirradiation

1. WAS THIS TISSUE MEANINGFULLY IRRADIATED?

Review the original plan, not only the prescription. A recurrent edge may have received far less than the nominal dose.

2. CAN ALL IMPORTANT DISEASE BE ENCOMPASSED?

A beautifully conformal target does not help if gross disease extends beyond a safely treatable volume.

3. WHICH ORGAN IS DOSE-LIMITING?

Carotid, mandible, temporal lobe, brainstem, optic apparatus, skin, and cranial nerves have different consequences and uncertainties.

4. WHAT OUTCOME WOULD JUSTIFY THE RISK?

Potential cure, durable control, symptom relief, preservation of airway/vision/nerve function, or prevention of bleeding.

CLINICAL PEARL | New facial weakness, numbness, trismus, or deep pain after treatment should be treated as recurrent perineural or skull-base disease until an adequate evaluation proves otherwise.

Chapter synthesis

- Complete salvage resection can be durable in selected isolated recurrences.
- Reirradiation requires cumulative-dose, geometry, and benefit analysis at an experienced center.
- Local treatment can remain valuable when systemic disease is present if it relieves symptoms or controls a threatening site.

CHAPTER 6

Distant Metastases: Observe, Ablate, or Treat

Metastatic urgency is determined by tempo and threat, not by the word “metastatic.”

The presence of metastasis does not determine its urgency. Histology, distribution, growth rate, symptoms, organ threat, and available local therapy do.

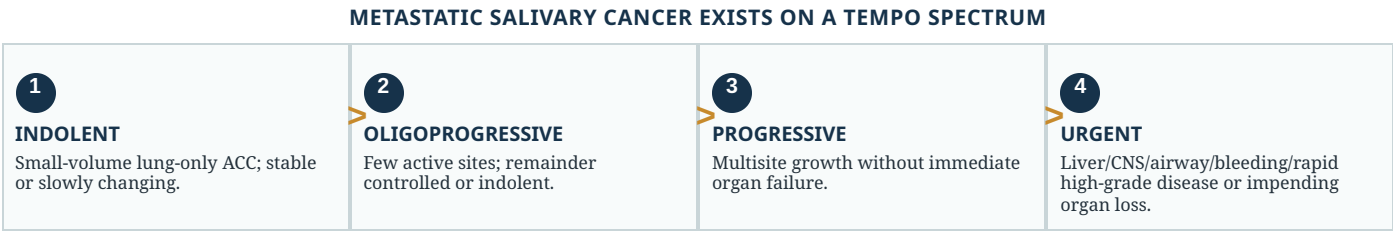


Figure 8. Original synthesis. Treat the patient’s trajectory, not the most frightening scan or the most reassuring histology label.

6.1 When observation is reasonable

ESMO supports watchful observation for selected ACC patients with lung-only metastases, low burden, no symptoms, and stable disease. This requires reproducible measurements and an explicit trigger for intervention; it is not abandonment.^{[2][3]} Short-interval imaging can establish growth kinetics, after which the interval may be lengthened when the trajectory is convincingly indolent.

6.2 Triggers to start systemic treatment

Trigger	Why it matters
Consistent radiographic progression	Confirms that disease is biologically active rather than merely present.
Symptoms or declining function	Creates a patient-centered treatment target: pain, dyspnea, neurologic deficit, bleeding, fatigue, or weight loss.
Liver, bone, CNS, airway, or other organ-threatening distribution	Consequences are less forgiving than small pulmonary nodules.
Rapid high-grade or SDC behavior	Delays may permit abrupt loss of control; AR and HER2 testing should already be available.
No feasible local-control option	Systemic therapy becomes the main disease-modifying strategy.
Time-sensitive trial opportunity	Measurable progression and performance status may create an opportunity that is lost after further decline.

PATIENT COUNSELING | “We are not deciding whether the metastases are real; we are deciding whether treatment now is more likely to help than to expose you to toxicity without changing what you feel or how the disease behaves.”

CHAPTER 6 CONTINUED

Local treatment, active observation, and concurrent palliative care

6.3 Oligometastatic and oligoprogressive disease

Surgery, stereotactic radiation, or ablation may be considered for selected limited metastases, especially after a long disease-free interval and when all active disease can be treated. In ACC, retrospective pulmonary-metastasectomy data suggest the most favorable outcomes when complete resection is possible and the disease-free interval is long, often cited as more than approximately 36 months.^[2] Local therapy can also address a painful bone lesion or a small number of progressing sites while systemic therapy controls the remainder.

OLIGOMETASTATIC

A limited number of sites at first metastatic presentation. Consider whether all visible disease can be treated and whether occult disease is likely to dominate soon.

OLIGOPROGRESSIVE

A few resistant sites progress while the remainder remains controlled. Local therapy may preserve benefit from an otherwise useful systemic regimen.

SYMPTOM-DIRECTED

A single site may merit radiation, surgery, embolization, or stabilization even when total disease burden is not limited.

6.4 Palliative care is concurrent care

Palliative medicine should be integrated when symptoms, treatment burden, or prognostic uncertainty become substantial. It is not limited to end-of-life care. Pain, airway and bleeding planning, nutrition, fatigue, mood, caregiver burden, work, and advance-care preferences may improve both quality of life and the patient’s ability to receive disease-directed treatment.

6.5 Active observation must have a protocol

Element	Document
Baseline	Number, size, and location of lesions; consistent CT technique and measurement method.
Initial interval	Often short enough to establish growth kinetics and exclude rapid behavior.
Clinical monitoring	Symptoms, performance status, organ function, and sites at disproportionate risk.
Treatment trigger	RECIST-like progression, accelerating growth, symptoms, organ threat, local-treatment opportunity, or trial window.
Exit plan	Which systemic family or local therapy becomes preferred when the trigger is reached.

Chapter synthesis

- Observation is legitimate for selected stable lung-only ACC when treatment triggers are explicit.
- Local ablative therapy may control limited metastatic or resistant sites in carefully selected patients.
- Palliative care should run alongside, not after, disease-directed treatment.

CHAPTER 7

A Systemic-Therapy Framework

Treat measurable biology with a defined goal - not a tumor name in isolation.

The treatment name should come after the histology, biomarker, clinical trajectory, and therapeutic goal are clear.

ADVANCED SALIVARY CANCER: HISTOLOGY AND BIOLOGY CHOOSE THE TREATMENT FAMILY

1 CONFIRM NEED Progression, symptoms, organ threat, or no useful local option.	>	2 CONFIRM DIAGNOSIS Re-review pathology and exclude high-grade transformation or metastatic mimic.	>	3 TEST INTELLIGENTLY AR/HER2 for SDC; fusion assay for secretory carcinoma; DNA + RNA when appropriate.	>	4 DEFINE GOAL Shrinkage, symptom relief, stabilization, bridge to local therapy, or trial access.	>	5 MATCH TOXICITY Cardiac/pulmonary, hormonal, vascular, renal, neurologic, and marrow risk.
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7.1 Before ordering systemic therapy

1. Confirm that systemic treatment is actually indicated.
2. Review the primary and recurrent pathology; behavior discordant with the label should trigger re-review or re-biopsy.
3. Obtain AR and HER2 testing for SDC and related high-grade adenocarcinomas; confirm an NTRK fusion in advanced secretory carcinoma.
4. Use broad next-generation sequencing when no standard biomarker explains active disease; include RNA-based fusion detection when relevant.
5. State the treatment goal and the clinical event that will determine whether therapy is working.
6. Assess comorbidity and toxicity before choosing among otherwise plausible options.

7.2 Tissue strategy

USE ARCHIVED TISSUE WHEN • Diagnosis and biomarker are secure • Material is not decalcified or exhausted • Disease phenotype is unchanged • Assay is technically appropriate	RE-BIOPSY WHEN • High-grade transformation is possible • Old FNA/cell block is scant • Decalcification may cause false negatives • A modern high-impact therapy depends on the result
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TUMOR-BOARD QUESTION | “What result would make us choose a different drug?” If the answer is unclear, the testing plan or the therapeutic question is not yet sufficiently specific.

CHAPTER 7 CONTINUED

Testing pitfalls and biomarker overlap

7.3 Molecular testing: common errors

Error	Why it fails	Better approach
Assuming secretory morphology proves an NTRK fusion	Mimics exist and targeted therapy depends on a real fusion	Confirm with a validated fusion assay, often RNA-based.
Using HER2 1+ or equivocal staining as if it were HER2-amplified SDC	Evidence and labeling differ by method and threshold	Report IHC score, ISH/amplification, assay, and heterogeneity.
Calling any AR staining “AR-driven”	Trials often enrich for strong/diffuse expression; thresholds vary	Document percentage and intensity and interpret with disease-specific expertise.
Relying on tiny, decalcified, or exhausted tissue	False negatives and incomplete profiling result	Re-biopsy a safe active site when the result will change therapy.
Ordering DNA-only testing in a fusion-driven tumor	Some rearrangements are missed by panel design or intronic coverage	Use RNA sequencing or validated fusion testing when appropriate.
Treating a target without considering tempo	An alteration determines drug choice, not automatically treatment timing	Integrate biology with symptoms, growth rate, local options, and goals.

7.4 When AR and HER2 overlap

Many SDCs express AR, and a clinically important subset overexpress or amplify HER2. There is no universally proven sequence when both are present. The choice should integrate the need for rapid shrinkage, strength and uniformity of HER2 expression, AR intensity, prior therapy, organ function, toxicity, and trial availability.

FAVOR AN ACTIVE HER2 REGIMEN WHEN

- Rapid visceral progression or high shrinkage need
- Strong, convincing HER2 positivity
- Cardiac and marrow reserve are adequate
- Patient accepts taxane or ADC toxicity

FAVOR ANDROGEN BLOCKADE WHEN

- Strong diffuse AR with slower disease tempo
- Desire to avoid cytotoxic chemotherapy
- Cardiac/pulmonary issues complicate HER2 therapy
- Hormonal adverse effects are acceptable

EVIDENCE LIMITATION | This is a clinical sequencing framework, not a randomized comparison. Reassess the molecular phenotype at progression when a new biopsy is feasible and management-changing.

Chapter synthesis

- Advanced therapy is selected by histology, biomarker, tempo, and treatment goal.
- RNA-based testing may be necessary for fusion-driven tumors.
- Actionable biology informs drug choice but does not determine when treatment must begin.
- Overlapping AR and HER2 require individualized sequencing rather than a reflex hierarchy.

CHAPTER 8

Histology-Directed Therapy: AR-Positive SDC

Biologic selection matters, and regimens within the same pathway are not equivalent.

Recurrent/metastatic SDC is often clinically aggressive and should be profiled early for AR and HER2.

8.1 Androgen-receptor-directed therapy

COMBINED ANDROGEN BLOCKADE Leuprorelin + bicalutamide Prospective phase II, 36 patients ORR 41.7% Clinical benefit 75% Median PFS 8.8 months	ENZALUTAMIDE MONOTHERAPY Alliance A091404, 46 patients Confirmed ORR 4.3% Median PFS 5.6 months Limited response durability	OTHER AR STRATEGIES Apalutamide-goserelin and abiraterone-based approaches provide additional phase II signals, including after prior androgen deprivation, but are not interchangeable with the strongest first-line evidence.
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Combined androgen blockade has the clearest prospective activity in strongly AR-positive SDC and related adenocarcinoma.^[6] Enzalutamide monotherapy produced few confirmed responses despite tumor regression in some patients, demonstrating that all androgen-pathway regimens are not equivalent.^[7]

Practical selection and toxicity

Issue	Clinical implication
AR interpretation	Record nuclear staining percentage and intensity; “positive” is not a universal binary threshold.
Tempo	Hormonal therapy may be attractive in slower disease; organ-threatening progression may require a regimen with a higher probability of rapid shrinkage.
Metabolic/sexual effects	Counsel regarding hot flashes, libido/erectile effects, fatigue, weight and metabolic changes.
Bone and cardiovascular health	Assess long-term androgen-deprivation consequences and manage modifiable risk.
Castration-resistant progression	Re-biopsy/sequence when feasible; consider another AR strategy, HER2 therapy if co-positive, chemotherapy, or trial.

BOARD PEARL | Do not answer “AR-positive SDC = enzalutamide.” The prospective combined-blockade data are substantially more compelling than enzalutamide monotherapy.

CHAPTER 8 CONTINUED

HER2-positive SDC

Separate disease-specific evidence, later-line signals, and current regulatory eligibility.

8.2 HER2-directed therapy

TRASTUZUMAB + DOCETAXEL Disease-specific phase II 57 HER2-positive SDC ORR 70.2% Median PFS 8.9 months Meaningful cytopenia and cardiac monitoring	TRASTUZUMAB + DOCETAXEL-PM Multicenter phase II 43 patients ORR 69.8% Better outcomes with IHC3+ or amplification than IHC2+ alone	T-DXD 17-patient salivary pooled analysis ORR 58.8% Median PFS 20.5 months Drug-related ILD 29.4% in this small cohort	T-DM1 / OTHER COMBINATIONS Lower-level evidence Small case series and later-line experience Do not place at the same evidence level as prospective trastuzumab-taxane studies.
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Trastuzumab plus docetaxel has the strongest disease-specific prospective evidence for HER2-positive SDC.^[5] A second multicenter phase II study reported a similar response rate and reinforced that IHC3+ or amplification predicts more convincing benefit than IHC2+ alone.^[8]

T-DXd has salivary-specific activity from a small pooled phase I analysis.^[9] Separately, the current US tumor-agnostic accelerated approval applies to adults with previously treated unresectable or metastatic **HER2-positive IHC3+** solid tumors without satisfactory alternatives.^[10] Do not confuse the broader “HER2-expressing” research cohort with the exact regulatory eligibility language.

Evidence hierarchy and safety

Option	How to frame it	Key safety issue
Trastuzumab-taxane	Preferred discussion for conventional strongly HER2-positive SDC when active treatment and shrinkage are needed	Cardiac function, cytopenias, neuropathy, infection, taxane effects
T-DXd	Later-line or labeled tissue-agnostic option in eligible IHC3+ disease; sequence depends on prior therapy and access	Interstitial lung disease/pneumonitis, nausea, marrow toxicity; stop and evaluate promptly for pulmonary symptoms
T-DM1	Promising later-line small-series evidence, not equivalent to a randomized or large prospective salivary trial	Cytopenia, hepatic toxicity, neuropathy
Dual HER2 combinations / trials	Reasonable in experienced centers or trials, especially after progression	Toxicity and evidence vary by regimen

DIAGNOSTIC TRAP | HER2 IHC2+ without amplification is not the same as uniformly IHC3+ or amplified SDC. Document assay, score, ISH result, and heterogeneity before committing to a high-impact treatment.

CHAPTER 8 CONTINUED

NTRK Fusion-Positive Secretory Carcinoma

A highly active treatment requires a real fusion and a real clinical indication.

8.3 Secretory carcinoma and NTRK fusion

Secretory carcinoma is the clearest precision-oncology success in salivary cancer. In 24 patients with TRK fusion-positive salivary cancers treated with larotrectinib, the objective response rate was 92%, 13% achieved complete response, and the 24-month PFS rate was 78%.^[11] This degree of activity justifies confirming the fusion in advanced disease rather than treating from morphology alone.

CONFIRM IDENTITY

- ETV6-NTRK3 is classic, but alternative fusions and morphologic mimics exist.
- Use validated fusion testing; RNA-based assays are often high yield.
- Do not rely on pan-TRK IHC alone as definitive proof.

CHOOSE AND MONITOR THERAPY

- Larotrectinib or entrectinib are established TRK-inhibitor concepts.
- Monitor hepatic and neurologic toxicity and drug interactions.
- At resistance, re-biopsy or plasma/tissue sequencing may identify a secondary kinase-domain mechanism and a trial option.

Treatment sequence



BOARD PEARL | The correct answer is not merely “secretory carcinoma = larotrectinib.” It is “advanced secretory carcinoma with a confirmed NTRK fusion and an indication for treatment = TRK inhibitor.”

8.4 Other fusion-defined tumors

Broader DNA and RNA profiling may reveal rare RET, ALK, BRAF, RAF-family, or other actionable events in tumors that lack a standard salivary treatment pathway. Match therapy should be based on assay quality, tissue-agnostic approval or trial evidence, disease tempo, and the patient’s need for treatment rather than on the novelty of the result alone.

CHAPTER 8 CONTINUED

Adenoid Cystic Carcinoma

Systemic therapy is chosen by progression and threat; disease stabilization can be meaningful.

8.5 Adenoid cystic carcinoma

Systemic therapy should generally be reserved for documented progression, symptoms, or threatening distribution. Cytotoxic response rates are modest. VEGFR-directed multikinase inhibitors commonly produce stabilization and occasional responses; no systemic therapy has established an overall-survival benefit.

Strategy	Evidence setting	Signal	How to counsel
Lenvatinib	Phase II, progressive R/M ACC	ORR 15.6%; stable disease 75%	Dose modification and discontinuation were common; hypertension, proteinuria, oral/mucosal and other VEGFR toxicities.
Axitinib	Phase II / randomized phase II	Low objective response, but PFS benefit versus observation in progressive disease	Activity is often stabilization rather than dramatic shrinkage.
Axitinib + avelumab	Phase II, progressive ACC	Confirmed ORR 18%; median PFS 7.3 months	Investigational; the incremental contribution of immunotherapy remains uncertain.
Amivantamab	2026 phase II, 18 evaluable patients	Clinical-benefit rate 72.2%	Emerging evidence; clinical benefit includes stable disease and does not establish a standard sequence.
NOTCH-directed / ADC / MYB trials	Biomarker-selected or early-phase studies	Biologic and antitumor signals	Trial referral is important, especially for aggressive or NOTCH-activated ACC.

Lenvatinib produced a 15.6% confirmed response rate with 75% stable disease in a progressive cohort.^[12] Axitinib has both single-arm and randomized data supporting disease control in progressive disease.^{[13][14]} Axitinib plus avelumab produced an 18% confirmed response rate but remains investigational.^[15]

EMERGING EVIDENCE - JULY 2026 | A small phase II amivantamab study reported a 72.2% clinical-benefit rate in recurrent/metastatic ACC. Window-of-opportunity AL101 data demonstrate biologic NOTCH-pathway modulation, and antibody-drug conjugate and MYB-directed trials are active. These are trial-level signals, not new universal standards.

ACC treatment priorities

INDOLENT LUNG-ONLY

Observe with reproducible imaging and explicit triggers when asymptomatic and stable.

PROGRESSIVE BUT NONURGENT

VEGFR inhibitor or trial; value stabilization and toxicity-adjusted time on therapy.

AGGRESSIVE / ORGAN-THREATENING

Trial referral, systemic treatment, and site-directed therapy according to molecular subtype and immediate threat.

CHAPTER 8 CONTINUED

Other Histologies, Chemotherapy, and Immunotherapy

8.6 Conventional chemotherapy

For progressive salivary carcinomas without a useful target, platinum-based therapy is a reasonable palliative option, but there is no universally preferred regimen. Carboplatin-paclitaxel is commonly used and produced a 39% response rate in a 38-patient retrospective cohort, with only 9% response in ACC.^[16] CAP - cyclophosphamide, doxorubicin, and cisplatin - has historical activity, but reports vary widely and the regimen adds anthracycline, renal, marrow, and emetic burden.^{[17][18]}

Clinical need	Reasonable discussion	Caveat
Rapid shrinkage without a target	Platinum-taxane or another histology-directed combination	Expect palliative rather than curative benefit; response varies markedly by histology.
Selected fit patient with need for combination therapy	CAP or another cisplatin-based regimen in experienced practice	Older, heterogeneous evidence and greater toxicity; define why it is preferable to a taxane regimen.
ACC with slow progression	VEGFR inhibitor or trial often preferred to reflex cytotoxic therapy	Cytotoxic response is usually modest; stable disease must be interpreted against pre-treatment growth.
Frailty or modest symptom burden	Single-agent or supportive/local therapy	Treatment burden may exceed benefit; state the patient-centered goal.

8.7 Immunotherapy

Checkpoint inhibitors produce durable responses in a small minority of unselected salivary cancers. KEYNOTE-158 enrolled 109 patients and reported an overall response rate of 4.6%; response was 10.7% in PD-L1-positive and 2.6% in PD-L1-negative tumors, and all five responses lasted at least two years.^[19] PD-L1 alone is therefore an imperfect selector.

MOST DEFENSIBLE USE

- Clinical trial
- MSI-high or mismatch-repair-deficient tumor
- High tumor mutational burden under an applicable tissue-agnostic indication
- Rare selected patient after standard options

DO NOT OVERSTATE

- PD-L1 positivity is not a strong salivary-specific predictor
- Low response rate does not mean zero chance of durable benefit
- VEGFR-checkpoint combinations remain investigational

Chapter synthesis

- Biomarker-selected AR, HER2, and NTRK therapies have the strongest systemic signals.
- ACC therapy often aims for disease control; trial referral is central.
- Conventional chemotherapy is palliative and histology-dependent.
- Unselected immunotherapy has a low response rate but occasional durable responses.

CHAPTER 9

Surveillance and Detection of Late Failure

The schedule should establish trajectory and preserve a window for salvage.

A surveillance plan should be reproducible enough to establish trajectory and long enough to respect salivary tumor biology.

SURVEILLANCE MUST MATCH BOTH EARLY RISK AND LATE BIOLOGY

Interval	Core plan	Purpose and caveat
~3 months after treatment	Contrast CT or MRI of primary site; chest imaging; PET/CT selectively	Establish postoperative/post-RT baseline and distinguish expected treatment change from disease.
Years 0-2	History/exam regularly; primary-site and chest imaging every 6-12 months	Highest generic early recurrence intensity; shorter interval for high-grade or suspicious findings.
Years 3-5	Annual or symptom/risk-directed primary-site and chest imaging	Individualize according to histology, grade, PNI, prior findings, and salvageability.
Beyond 5 years	At least annual examination; ongoing chest or neural imaging for selected higher-risk histologies	Late recurrence is clinically important in ACC and some other salivary cancers.

ASCO recommends regular history and examination with decreasing frequency, baseline imaging at approximately three months, primary-site and chest imaging every 6-12 months during the first two years, and risk- or symptom-directed imaging during years three through five. Annual examination continues beyond five years, with longer-term chest imaging considered in higher-risk disease.^[1]

9.1 What the visit should detect

Domain	Ask / examine	Concerning change
Primary bed	New mass, skin change, wound breakdown, deep pain, trismus, mucosal lesion	Progressive focal finding rather than stable postoperative asymmetry
Cranial nerves	Facial movement/sensation, tongue, palate, voice, shoulder as relevant	New weakness, numbness, neuralgia, dysphonia, denervation symptoms
Neck	Parotid bed and levels; scalp/auricle/face when cutaneous metastasis was relevant	New firm node or intraparotid mass
Chest/systemic	Cough, dyspnea, bone pain, weight loss, neurologic symptoms	New symptom in a histology with distant risk
Treatment toxicity	Dry mouth, dental disease, swallow, eye closure, hearing, edema, pain, mood	Functional decline that is treatable even without recurrence

CHAPTER 9 CONTINUED

ACC and metastatic-disease surveillance

9.2 Imaging choices

Contrast MRI is particularly useful for deep soft tissue, skull base, denervation, and perineural recurrence. CT is efficient for cortical bone, nodes, and pairing local imaging with chest surveillance. PET/CT is useful selectively in high-grade or unclear recurrence but is not a universal surveillance test. Use a consistent protocol whenever possible so growth kinetics remain interpretable.

9.3 ACC-specific horizon

ESMO **suggests** contrast head-and-neck MRI every three to four months during the first two years after ACC treatment, every six months during years three through five, and annually thereafter, together with at least annual lung CT.^{[2][3]} This is low-level consensus guidance, not a trial-proven schedule. Modify it for age, prior findings, treatment burden, symptoms, and whether detecting a recurrence would change management.

EVIDENCE LANGUAGE | Prefer “ACC warrants prolonged surveillance” or “ESMO suggests” over “ACC requires” a single universal schedule. The clinical rationale is strong; the exact interval is consensus-based.

9.4 Surveillance after known metastatic disease

Clinical state	Imaging objective	Reasonable interval concept
Newly recognized small lung-only ACC metastases	Establish growth kinetics	Short interval initially, then lengthen if convincingly indolent
Active disease on systemic therapy	Measure response and detect toxicity/organ threat	Protocol- or regimen-directed; usually more frequent early
Oligoprogressive disease	Identify whether only a few sites are escaping control	Interval sufficient to preserve a local-treatment window
No treatment option that would change with imaging	Reconsider the purpose of repeated scans	Avoid surveillance that adds burden without changing decisions

9.5 The post-treatment baseline is an anatomic record

The baseline study should document the flap, deep-lobe or parapharyngeal space, facial-nerve pathway, skull base, neck dissection bed, and chest. Later imaging should be compared directly with this study, not only with the immediately preceding scan.

PATIENT COUNSELING | “Follow-up becomes less frequent, but it does not end at five years. Some salivary cancers recur late, so yearly review and a histology-specific chest or neural imaging plan remain important.”

Chapter synthesis

- Use baseline imaging for longitudinal comparison.
- Imaging intensity is highest early, then becomes histology- and risk-directed.
- ACC deserves a prolonged horizon, but the exact schedule remains consensus-based.
- Surveillance visits must detect treatable toxicity as well as recurrence.

CHAPTER 10

Survivorship and Rehabilitation

Long-term care is an active multidisciplinary treatment program.

Cancer control is incomplete when the patient is left with preventable visual, dental, swallowing, hearing, or facial disability.

ORAL HEALTH

Dental evaluation before meaningful oral/mandibular dose; lifelong fluoride, hygiene, diet counseling, saliva support, and dose-aware extractions.

SWALLOW AND TRISMUS

SLP before/during treatment when risk is meaningful; maintain safe oral intake and jaw range before disuse and fibrosis are established.

FACIAL NERVE AND EYE

Serial corneal protection, resting tone, synkinesis, reinnervation timeline, facial therapy, and candidacy for dynamic/static procedures.

HEARING/VESTIBULAR

Cochlear dose, skull-base field, ear surgery, and ototoxic drugs determine risk; use audiometry when the result changes rehabilitation.

FIBROSIS/LYMPHEDEMA

Neck and facial edema, range limitation, and swallow effects can respond to specialized therapy; progressive change also requires recurrence evaluation.

PAIN/PSYCHOSOCIAL

Neuropathic pain, sleep, mood, work, finances, caregiver burden, and uncertainty deserve active treatment and palliative support when appropriate.

10.1 Xerostomia and dental prevention

Tissue-sparing RT should be used when feasible. For established salivary hypofunction, topical lubricants, saliva substitutes, sugar-free lozenges or gum, and selected sialogogues can help when residual gland function exists.^{[20][21]} The patient should understand that xerostomia is a chronic medical condition with caries, infection, nutrition, sleep, and quality-of-life consequences.

10.2 Dysphagia, trismus, fibrosis, and lymphedema

Maintain swallowing and jaw movement when safe rather than waiting for severe dysfunction. New progressive dysphagia, trismus, neuropathy, or deep pain years later requires evaluation for both late toxicity and recurrence.

10.3 Facial nerve and eye

Document what repair, graft, transfer, or static procedure was performed and the expected reinnervation horizon. A clear cornea today does not eliminate future exposure risk as tone, synkinesis, and eyelid position evolve.

CHAPTER 10 CONTINUED

Systemic toxicity and late-effect surveillance

10.4 Systemic-therapy survivorship

Survivorship extends beyond surgery and RT. Long disease control with targeted therapy can create chronic toxicities that require longitudinal prevention and rehabilitation.

Therapy family	Important late or cumulative effects	Surveillance priorities
Androgen blockade	Hot flashes, fatigue, sexual dysfunction, metabolic change, bone loss, cardiovascular risk, hepatic effects	Baseline metabolic/bone risk; symptom counseling; appropriate bone and cardiovascular prevention
Trastuzumab-taxane	Cardiac dysfunction, cytopenia, neuropathy, infection, fatigue	Cardiac assessment and monitoring; marrow and neuropathy surveillance
T-DXd	ILD/pneumonitis, nausea, cytopenia, fatigue	Immediate evaluation of cough/dyspnea/fever; hold therapy and follow current labeling/oncology protocols
VEGFR inhibitors	Hypertension, proteinuria, bleeding, mucosal toxicity, fatigue, thyroid dysfunction, wound-healing effects	Blood pressure, urinalysis/protein, thyroid and bleeding review; coordinate procedures and dose holds
TRK inhibitors	Hepatic, neurologic, weight and drug-interaction effects	Liver tests, medication review, neurologic symptoms, resistance assessment
Checkpoint inhibitors	Immune endocrine, pulmonary, GI, hepatic, skin, neurologic toxicity	Recognize immune patterns early and follow organ-specific management pathways

10.5 Additional radiation-related late effects

ENDOCRINE AND VASCULAR

- Hypothyroidism after lower-neck exposure
- Pituitary dysfunction after selected skull-base fields
- Carotid and vascular late effects
- Skin and soft-tissue fibrosis

BONE AND DENTAL PROCEDURES

- Review mandibular/maxillary dose before extraction or implant planning
- Coordinate oral surgery and radiation oncology
- Do not estimate osteoradionecrosis risk from elapsed time alone

10.6 Functional decline is not automatically recurrence - or automatically toxicity

A new symptom should trigger a paired assessment: is this a treatable treatment effect, recurrent tumor, or both? Stable postoperative facial asymmetry differs from progressive branch weakness; chronic dry mouth differs from new focal jaw pain; baseline numbness differs from expanding neuralgia.

RESIDENT TAKEAWAY | At every follow-up visit, ask both questions: “Is the cancer recurring?” and “Is treatment causing a preventable or reversible disability?”

Chapter synthesis

- Dental, swallowing, facial, ocular, hearing, and lymphedema prevention begins before toxicity is established.
- Systemic therapies create their own chronic survivorship needs.
- New late neurologic, pain, or functional symptoms require oncologic evaluation as well as rehabilitation.

CHAPTER 11

Integrated Clinical Cases

Postoperative radiation and local-regional reasoning

The correct answer follows from failure pattern, tempo, resectability, biomarker, and evidence level - not from memorizing one treatment name.

Case 1 - Small ACC with focal microscopic PNI	
A 42-year-old has a 1.4-cm hard-palate ACC resected with a confidently mapped negative margin and no nodal disease. There is focal microscopic PNI. Questions: Is PORT indicated? Which nerve route matters? Is the recommendation universal?	Answer PORT is strongly supported under ASCO, and focal PNI adds a specific neural-risk rationale. Map the palatal nerve toward the pterygopalatine fossa/V2 as appropriate. Acknowledge that ESMO uses a more adverse-feature-directed framework; this patient still has PNI and would generally be recommended PORT.
Case 2 - Early low-grade parotid carcinoma	
A 55-year-old has a 1.8-cm low-grade acinic cell carcinoma removed intact with a mapped negative margin, no LVI/PNI, intact facial nerve, and negative neck. Questions: Is PORT mandatory? What must be confirmed before observation?	Answer Observation is reasonable. Confirm that the entity, grade, margin, and extent are secure and not inferred from a limited sample. Document a baseline and a histology-appropriate surveillance plan.
Case 3 - Positive margin, ENE, and HER2 amplification	
A high-grade SDC has a focally positive facial-nerve interface and two level II nodes with ENE. The tumor is strongly HER2-positive. Questions: Re-resection? PORT? Cisplatin? Adjuvant trastuzumab?	Answer Discuss re-resection if the mapped margin can be cleared without disproportionate morbidity. PORT to the bed and nodal levels is strongly indicated. Concurrent cisplatin is not routine outside a trial. HER2-directed therapy treats measurable advanced disease and is not routine adjuvant therapy after complete local treatment outside a trial.
ORAL-BOARD STRUCTURE State the final pathology, failure map, resectability of any residual disease, PORT indication, exact target, treatment timing, baseline imaging, and the function-specific follow-up plan.	

CHAPTER 11 CONTINUED

Metastatic tempo and biomarker sequencing

Case 4 - Stable pulmonary ACC metastases

Eight years after ACC treatment, six bilateral lung nodules measuring 4-9 mm have changed minimally over 18 months. The patient is asymptomatic, with no locoregional disease.

Questions: Must systemic treatment start? What makes observation active rather than passive?

Answer

No. Continue active surveillance with reproducible measurements and explicit triggers: accelerating growth, symptoms, organ threat, local-treatment opportunity, or trial window. Metastasis alone is not an indication for toxic therapy in indolent ACC.

Case 5 - Rapidly progressive AR-positive/HER2-negative SDC

The patient develops rapidly enlarging pulmonary and hepatic metastases. Tumor cells show strong diffuse AR expression and no convincing HER2 overexpression or amplification.

Questions: Observe? Enzalutamide? Combined blockade? Cytotoxic therapy?

Answer

This is active organ-threatening disease and should be treated promptly. Combined androgen blockade has substantially stronger prospective response data than enzalutamide monotherapy. A high need for rapid shrinkage, hepatic function, trial access, and cytotoxic options should be discussed.

Case 6 - AR/HER2 co-positive SDC

A fit patient has diffuse AR expression, HER2 IHC3+, and rapidly progressive visceral disease.

Questions: Which target should be treated first? Is there a proven sequence?

Answer

No randomized sequence is established. Rapid visceral progression and strong HER2 positivity often favor an active HER2-directed regimen because of the high response probability. Combined androgen blockade remains a plausible line of therapy. State the goal, toxicity tradeoffs, and reassessment plan rather than claiming one universal order.

BOARD TRAP | A biomarker chooses a treatment family. It does not by itself decide urgency, sequence, or whether systemic treatment is needed at all.

CHAPTER 11 CONTINUED

Salvage, reirradiation, and neural recurrence

Case 7 - Secretory carcinoma with a confirmed NTRK fusion

Unresectable recurrent secretory carcinoma carries an ETV6-NTRK3 fusion. The patient has symptomatic progressive disease.

Questions: Preferred systemic concept? Is morphology enough? What is the resistance plan?

Answer

Use a TRK inhibitor such as larotrectinib or entrectinib. Fusion confirmation matters because treatment is biomarker-defined. At progression, obtain tissue or molecular analysis when feasible to identify resistance and a next-generation inhibitor trial.

Case 8 - Small recurrence in a previously irradiated parotid bed

Seven years after surgery and PORT, a 1.2-cm focal recurrence abuts the masseter. Chest imaging is clear and the facial nerve remains functional.

Questions: Salvage surgery? Reirradiation? What information is required?

Answer

Biopsy and map the recurrence. If complete salvage resection is feasible with acceptable nerve cost, surgery offers the clearest chance of durable control. Obtain the original RT plan before considering postoperative or definitive reirradiation; review composite dose, target geometry, and expected benefit.

Case 9 - New V2 symptoms after treatment

Three years after treatment of a palatal ACC, the patient develops progressive cheek numbness and deep facial pain. The oral cavity and neck examination are normal.

Questions: What study? What route? Does the normal superficial examination reassure you?

Answer

Urgently obtain contrast MRI along the palatal/pterygopalatine/V2 pathway toward foramen rotundum and review with neuroradiology. A normal mucosal or neck examination does not exclude named-nerve recurrence.

CLINICAL BRIDGE | When a recurrence question will determine a morbid operation, high-risk reirradiation, or modern targeted therapy, tissue confirmation and multidisciplinary image review become especially valuable.

CHAPTER 11

Boards-Style Consolidation

High-yield principles with the important caveats preserved.

A strong written- or oral-board answer states the evidence level, not only the treatment name.

Prompt	Best principle
Resected ACC	ASCO: offer PORT at all stages. ESMO: feature-directed; discuss the difference explicitly.
High grade, positive margin, PNI, nodal metastasis, LVI, T3-T4	Offer PORT.
Close margin or intermediate grade alone	Individualize by anatomy and coexisting risk.
PORT timing	Aim to begin within eight weeks while completing wound, dental, nutritional, functional, and simulation preparation.
Named-nerve PNI/spread	Map the associated nerve to the relevant skull-base corridor; distinguish microscopic PNI from gross spread.
Concurrent cisplatin with PORT	Not routine outside a trial; RTOG 1008 results remain pending as of July 29, 2026.
Unresectable local disease	Definitive RT; concurrent systemic radiosensitization remains unproven.
Stable asymptomatic lung-only ACC metastases	Observation can be appropriate with explicit triggers.
AR-positive progressive SDC	Combined androgen blockade has prospective activity; monotherapy evidence is weaker.
HER2-positive SDC	Trastuzumab-taxane has strong disease-specific prospective activity; T-DXd and T-DM1 occupy different evidence/regulatory tiers.
NTRK fusion secretory carcinoma	TRK inhibitor when treatment is indicated.
Unselected immunotherapy	Low ORR; use trial or established tissue-agnostic biomarker when possible.
Surveillance beyond five years	Annual examination; continue chest/neural imaging selectively according to histology and risk.

Failure-pattern language

LOCAL BED Margin, extension, reconstruction, residual disease	NERVE Peripheral branch, named nerve, skull-base corridor	REGIONAL Intraparotid nodes, levels, ENE, next echelon	DISTANT Histology, tempo, distribution, symptoms, organ threat
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BOARDS LANGUAGE | Do not say only “I will discuss at tumor board.” State the pathology feature, the likely failure route, what treatment is being recommended, what uncertainty remains, and what finding would change the plan.

CHAPTER 12

Resident Workflow and One-Page Synthesis

A reproducible checklist protects against omission when pathways are rare.

Postoperative tumor-board card

Question	Write the answer
What was resected?	Primary site, T category, adjacent structures, nerves, neck levels, reconstruction.
What is the final biology?	Histology, grade, high-grade transformation, molecular phenotype.
Where is residual risk?	Mapped positive/close margin, PNI/named nerve, LVI, nodes/ENE, intentional residual disease.
Is PORT indicated?	ASCO/ESMO framework; high grade, margin, PNI, nodes, LVI, T3-T4; individualized close/intermediate grade.
What must be targeted?	Bed, named interface, involved nodes, next echelon, nerve to skull base, gross residual.
Can RT begin by eight weeks?	Wound, dental, nutrition, SLP, hearing, simulation, referrals.
What is the baseline?	Post-treatment CT/MRI at about three months; chest baseline; PET/CT selectively.
What is the surveillance horizon?	Histology, grade, neural risk, chest risk, late-recurrence plan.
What tissue is preserved?	AR, HER2, NTRK, DNA/RNA testing as relevant.
What function requires follow-up?	Eye/facial nerve, dental/xerostomia, swallow, hearing, endocrine, lymphedema, pain, psychosocial health.

The five decisions to say out loud

1 FAILURE MAP Local bed, named nerve, nodes, distant risk, molecular options.	>	2 ADJUVANT CHOICE Observation, re-resection, PORT, or trial - and why.	>	3 TARGET MAP Exact bed, interface, nerve route, and nodal levels.	>	4 SURVEILLANCE Baseline, first two years, years 3-5, and long-term horizon.	>	5 RECURRENCE PLAN What triggers biopsy, salvage, ablation, systemic therapy, or a trial?
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FINAL RESIDENT TAKEAWAY | Do not let a rare histology make the plan vague. Every case can still be organized by anatomy, failure pattern, tempo, resectability, evidence level, and actionable biology.

Educational use

This guide supports resident education and multidisciplinary discussion. It does not replace current guidelines, institutional protocols, radiation planning, drug labeling, or patient-specific judgment.

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Guidelines, pivotal studies, and current evidence

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Publication notes and final framework

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31. CAP and taxane-based chemotherapy studies are small and histologically heterogeneous; no cytotoxic regimen has been established as a universal standard. See references 16-18 and the 2026 systemic-therapy review.

32. Clinical algorithms in this guide are educational syntheses and require multidisciplinary review, current drug labeling, and institution-specific radiation constraints.

Figure credits

Figures 1-4 and 6-13 are original educational syntheses created for the ENT Resident Library. Figure 5 uses an open-license MRI example of facial-nerve perineural tumor spread by Jto410, CC BY-SA 3.0, via Wikimedia Commons. All algorithms are educational summaries and do not replace current guidelines, institutional protocols, drug labeling, or patient-specific judgment.

FINAL RESIDENT TAKEAWAY | After treatment, keep five decisions explicit: failure map, adjuvant choice, target map, surveillance plan, and recurrence trigger.